D210 Data Dashboard and Storytelling

D210 Performance Assessment

Representation and Reporting
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A1: Interactive Tableau Dashboard

My dashboards are visible as part of a four part “Story” and are widely accessible at Tableau Public via the following link: D210 Data Dashboard and Storytelling

A2: Data Sets for Dashboard

The dashboard references two datasets: the cleaned medical dataset from WGU and a publicly available dataset from the Behavioral Risk Factor Surveillance System (BRFSS) survey conducted by the CDC in 2020. The BRFSS survey is the largest continuously conducted health survey system in the world, completing more than 400,000 adult interviews every year to assess the health status of U.S. residents across all 50 states as well as the District of Columbia and three U.S. territories (Kumar, 2025). Of the 279 original attributes, I used six variables that I could compare to variables within the WGU dataset: BMI, Stroke, Sex, AgeCategory, Diabetic, and Asthma.

To properly compare the two datasets, I still needed to conduct some level of data transformation prior to uploading them to Tableau. The main problematic variables were BMI and AgeCategory, as WGU only had a binary variable for Overweight and a continuous variable for Age. As such, I converted all BMI values in the CDC dataset over 24.9 to “Yes” and all values equal to or under 24.9 to “No” and renamed the variable “overweight” to match the dataset from WGU. For the issue of the Age variable, I transformed the WGU dataset to bin all ages to match the bins demonstrated in the CDC dataset, e.g. 18-24, 25-39, 40-44, etc., and including all ages equal to or above 80 in a single bin, 80 or Older. I also renamed all variables in the WGU dataset and the CDC set to match and follow proper naming conventions, e.g. lowercase and underscored where appropriate.

The CDC dataset was considerably larger than the one from WGU, with 401,958 rows compared to 10,000. As such, I sampled 10,000 random rows from the CDC dataset, setting the random_state generator to the standard value of 42 for reproducibility. Finally, I created an additional variable, “source”, which contained the value of “WGU” for the WGU dataset and “CDC” for the CDC dataset. In the end, I was left with two datasets with identical dimensions and variable names: source, gender, age_category, overweight, diabetes, asthma, and stroke. These datasets are attached as separate files. I then combined both datasets into a singular, combined dataset to facilitate dashboard creation in Tableau. This combined dataset is also attached as a separate file.

In [1]:
```
import pandas as pd
import numpy as np
file_path = r'C:\Users\My Name Devon\OneDrive\Documents\WGU MSDA\D210\medical_clean.csv'
df = pd.read_csv(file_path, index_col = 0)
df.head(5)
```

Out[1]:

CaseOrder	Customer_id	Interaction	UID	City	State	County	Zip	Lat	Lng
1	C412403	8cd49b13-f45a-4b47-a2bd-173ffa932c2f	3a83ddb66e2ae73798bdf1d705dc0932	Eva	AL	Morgan	35621	34.34960	-86.7250
2	Z919181	d2450b70-0337-4406-bdbb-bc1037f1734c	176354c5eef714957d486009feabf195	Marianna	FL	Jackson	32446	30.84513	-85.2290
3	F995323	a2057123-abf5-4a2c-abad-8ffe33512562	e19a0fa00aeda885b8a436757e889bc9	Sioux Falls	SD	Minnehaha	57110	43.54321	-96.6377
4	A879973	1dec528d-eb34-4079-adce-0d7a40e82205	cd17d7b6d152cb6f23957346d11c3f07	New Richland	MN	Waseca	56072	43.89744	-93.5147
5	C544523	5885f56b-d6da-43a3-8760-83583af94266	d2f0425877b10ed6bb381f3e2579424a	West Point	VA	King William	23181	37.59894	-76.8895

5 rows × 49 columns

In [2]:
```
bins = [0, 24, 29, 34, 39, 44, 49, 54, 59, 64, 69, 74, 79, float('inf')]
labels = ['18-24', '25-29', '30-34', '35-39', '40-44', '45-49', '50-54', '55-59', '60-64', '65-69', '70-74', '7
df['age_category'] = pd.cut(df['Age'], bins=bins, labels=labels, right=True)
print(df['age_category'].unique())
```

```
['50-54', '75-79', '18-24', '40-44', '45-49', ..., '70-74', '30-34', '35-39', '65-69', '25-29']
Length: 13
Categories (13, object): ['18-24' < '25-29' < '30-34' < '35-39' ... '65-69' < '70-74' < '75-79' < '80 or older']
```

In [11]:
```
df.rename(columns={"Stroke" : "stroke", "Gender" : "gender", "Diabetes" : "diabetes", "Asthma" : "asthma", "Ove
df = df.assign(source='WGU')
wgu_data = df[["source", "stroke", "gender", "age_category", "diabetes", "asthma", "overweight"]]
wgu_data.head(5)
```

Out[11]:

CaseOrder	source	stroke	gender	age_category	diabetes	asthma	overweight
1	WGU	No	Male	50-54	Yes	Yes	No
2	WGU	No	Female	50-54	No	No	Yes
3	WGU	No	Female	50-54	Yes	No	Yes
4	WGU	Yes	Male	75-79	No	Yes	No
5	WGU	No	Female	18-24	No	No	No

In [4]:
```
wgu_data.shape
```

Out[4]: (10000, 6)

In [5]:
```
# Importing new dataset from BRFSS
file_path_2 = r'C:\Users\My Name Devon\OneDrive\Documents\WGU MSDA\D210\heart_2020_cleaned.csv'
brfss_df = pd.read_csv(file_path_2)
brfss_df.head(5)
```

Out[5]:

	HeartDisease	BMI	Smoking	AlcoholDrinking	Stroke	PhysicalHealth	MentalHealth	DiffWalking	Sex	AgeCategory	Race
0	No	16.60	Yes	No	No	3	30	No	Female	55-59	White
1	No	20.34	No	No	Yes	0	0	No	Female	80 or older	White
2	No	26.58	Yes	No	No	20	30	No	Male	65-69	White
3	No	24.21	No	No	No	0	0	No	Female	75-79	White
4	No	23.71	No	No	No	28	0	Yes	Female	40-44	White

In [6]:
```
brfss_df = pd.read_csv(file_path_2)
brfss_df.info(memory_usage="deep")
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 319795 entries, 0 to 319794
Data columns (total 18 columns):
 #   Column            Non-Null Count   Dtype
---  ------            --------------   -----
 0   HeartDisease      319795 non-null  object
 1   BMI               319795 non-null  float64
 2   Smoking           319795 non-null  object
 3   AlcoholDrinking   319795 non-null  object
 4   Stroke            319795 non-null  object
 5   PhysicalHealth    319795 non-null  int64
 6   MentalHealth      319795 non-null  int64
 7   DiffWalking       319795 non-null  object
 8   Sex               319795 non-null  object
 9   AgeCategory       319795 non-null  object
 10  Race              319795 non-null  object
 11  Diabetic          319795 non-null  object
 12  PhysicalActivity  319795 non-null  object
 13  GenHealth         319795 non-null  object
 14  SleepTime         319795 non-null  int64
 15  Asthma            319795 non-null  object
 16  KidneyDisease     319795 non-null  object
 17  SkinCancer        319795 non-null  object
dtypes: float64(1), int64(3), object(14)
memory usage: 232.9 MB
```

In [7]:
```
bins = [0, 24.9, np.inf]
labels = ['No', 'Yes']
brfss_df['overweight'] = pd.cut(brfss_df['BMI'], bins=bins, labels=labels, right=True)
```

In [10]:
```
brfss_df.rename(columns={"Stroke" : "stroke", "Sex" : "gender", "AgeCategory" : "age_category", "Diabetic" : "di
brfss_df = brfss_df.assign(source='CDC')
truncated_brfss = brfss_df[["source", "stroke", "gender", "age_category", "diabetes", "asthma", "overweight"]]
truncated_brfss.head(5)
```

Out[10]:

	source	stroke	gender	age_category	diabetes	asthma	overweight
0	CDC	No	Female	55-59	Yes	Yes	No
1	CDC	Yes	Female	80 or older	No	No	No
2	CDC	No	Male	65-69	Yes	Yes	Yes
3	CDC	No	Female	75-79	No	No	No
4	CDC	No	Female	40-44	No	No	No

In [12]:
```
truncated_brfss.shape
```

Out[12]:
```
(319795, 7)
```

In [17]:
```
num_records_to_sample = 10000
sampled_brfss = truncated_brfss.sample(n=num_records_to_sample, random_state=42)
new_brfss = sampled_brfss.reset_index(drop=True)
new_brfss.head(10)
```

Out[17]:

	source	stroke	gender	age_category	diabetes	asthma	overweight
0	CDC	No	Female	25-29	No	No	Yes
1	CDC	No	Female	30-34	No	No	No
2	CDC	No	Female	40-44	No	Yes	Yes
3	CDC	No	Female	65-69	No	No	Yes
4	CDC	No	Female	60-64	No	Yes	Yes
5	CDC	No	Female	80 or older	Yes	No	No
6	CDC	No	Male	35-39	No	No	Yes
7	CDC	No	Female	25-29	No	No	No
8	CDC	No	Male	65-69	No	No	No
9	CDC	No	Male	45-49	No	Yes	Yes

In [14]:
```
new_brfss.shape
```

Out[14]:
```
(10000, 7)
```

In [15]:
```
#Create and save new datasets as individual csv files
new_brfss.to_csv('final_BRFSS_clean.csv')
wgu_data.to_csv('final_WGU_clean.csv')
```

In [19]:
```
#Place both datasets into a single sheet based on matching index, CaseOrder
combined_df = pd.concat([wgu_data, new_brfss])
combined_df
```

Out[19]:

	source	stroke	gender	age_category	diabetes	asthma	overweight
1	WGU	No	Male	50-54	Yes	Yes	No
2	WGU	No	Female	50-54	No	No	Yes
3	WGU	No	Female	50-54	Yes	No	Yes
4	WGU	Yes	Male	75-79	No	Yes	No
5	WGU	No	Female	18-24	No	No	No
...	...	...	...	...	...	...	...
9995	CDC	No	Female	25-29	No	Yes	Yes
9996	CDC	No	Male	80 or older	No	Yes	Yes
9997	CDC	No	Male	60-64	No	No	Yes
9998	CDC	No	Male	30-34	No	No	Yes
9999	CDC	Yes	Male	75-79	No	No	Yes

20000 rows × 7 columns

In [20]:
```
#Create and save new combined dataset
combined_df.to_csv('combined_df_clean.csv', index = False)
```

A3: Dashboard Installation

No installation is required to access the dashboards. One may view them at Tableau Public via the following link: D210 Data Dashboard and Storytelling

If desired, the presentation may be downloaded directly from the Tableau website to a PDF or PowerPoint file using the icon in the upper right corner above the presentation labeled "Download", displayed as a downward arrow within a rectangle.

A4: Dashboard Navigation

The first dashboard provides an overview of the patient demographics from both datasets, using age and gender as distinctive categories for comparison. Age distribution is demonstrated as two bar charts, one for the CDC and one for WGU, and gender distribution is demonstrated as two pie charts, again, one for the CDC and one for WGU. Three side filters allow the user to easily filter the results to narrow to one or multiple results for either gender, age category, or source. These filters apply to all charts on the first dashboard.

The second dashboard contains four separate heat maps for the four different common conditions between the WGU and CDC datasets: asthma, overweight, diabetes, and stroke. Each heat map has a corresponding legend that shows the minimum and maximum values of the color range. A table above each legend also shows the occurrence of each condition in the respective datasets as a percentage.

Finally, the third dashboard contains a succinct text summary, without charts or visualizations, that summarizes the data presented. This serves to aid audience members who may be less inclined to interpret visual cues alone and prefer to also benefit from a readable summary.

B1: Panopto Presentation

My Panopto Presentation may be viewed at the following link: Panopto Presentation

C1: Dashboard Purpose and Function

My dashboard's purpose is to support executive decision making by enabling leaders to explore the data, identify trends, and compare key metrics. The data dictionary explains that the Senior Vice President of Hospital Operations (SVP) is specifically interested in the demographic distribution of patient treatment and outcomes and how this may apply to patient care to guide strategic improvements in the field. I designed my dashboard with these priorities in mind, attempting to provide a clear understanding of the demographic distribution of patient treatment and outcomes within WGU and to compare these trends to a broader national sample. To fulfill this, I chose the CDC's 2020 BRFSS survey as a comparative dataset, as it offers publicly available data on patient age and gender in relation to conditions also found in the WGU dataset: asthma, diabetes, overweight, and stroke. Comparing WGU's internal data to this national dataset allows executive leaders to assess whether observed trends at WGU are unique or reflective of broader societal patterns.

C2: Insight from Public Dataset

The CDC's BRFSS dataset enhances the insights that can be drawn from the WGU dataset by providing a benchmark against which to evaluate WGU's patient population. Without the BRFSS data, we can only observe the demographic distribution of conditions within the

WGU dataset without any external context regarding national trends. For example, if the WGU dataset shows a high prevalence of diabetes among women aged 55-64, we can compare it to the BRFSS data to determine whether this is within ranges seen in the rest of the country or else unusually high or low. If WGU's prevalence is significantly higher, it might indicate the need for targeted interventions, such as diabetes screening programs or educational initiatives specifically designed for this demographic.

The demographic variables of age and gender allow us to examine the relationship between those factors and specific health conditions in a much broader context. We can assess whether the age-related increase in stroke prevalence observed at WGU is consistent with national trends, or whether WGU experiences a more pronounced increase at a particular age range, which might indicate specific risk factors or gaps in preventative care within their patient population.

C3: Executive Decision-Making

I designed my dashboard to address the SVP's interest in patient outcome as a function of demographic factors. As such, I used the first tab of my Tableau story to display the demographic distributions of my two cleaned datasets. Two bar charts demonstrate the age category distribution for both the CDC and WGU data, while two pie charts show the gender distribution for each. The following section features heat maps for each condition across age categories for both data sources. A percentage occurrence chart is included for each pair of heat maps, showing the upper and lower limits for each range. These heat maps visually highlight age-related trends in condition prevalence and can immediately indicate similarities and differences between trends at WGU and trends seen at the CDC level.

C4: Interactive Controls

Three interactive filters (age category, gender, and data source) are embedded across all the data visualizations, empowering users to explore specific subgroups and compare them across datasets. This allows executives to conduct more target analyses of the data presented. For example, the SVP can filter to female patients aged 55-64 and see if their prevalence of diabetes differs significantly between WGU and the national average, potentially highlighting areas where WGU can improve targeted care.

C5: Colorblindness Adaptations

The Tableau website includes an article with helpful tips on how to create dashboards that are friendly to colorblind users. While a relatively small percentage of the population is estimated to suffer from color vision deficiency (CVD) – about 8% of men and 0.5% of women - it is still best practice to ensure that products are accessible to as broad a population as possible. The most expedient method of catering to colorblindness is to avoid using colors associated with CVD – predominantly red and green. Blue and yellow CVD also occurs, but with an even smaller fraction of the population. The most common palette used for colorblind individuals focuses on contrasting blue and orange shades (Shaffer, 2016).

As such, I attempted to keep to these shades in most of my visualizations, using a slight variation of blue and red for the gender pie charts. The Color-Blind palette, already configured in Tableau, includes these shades, allowing me to apply it to my bar charts and heat maps automatically. I reversed the color range for the heat maps, as initially, blue was automatically associated with higher-density values, which I found visually confusing. I associate warm colors with higher numbers and cool colors with lower numbers. As such, I changed the color association to make dark orange the color associated with high density and dark blue the color associated with low density.

Beyond color, I also added a short introduction including details on the data sources, purpose, and design of the dashboards, as well as a textual summary at the end to address any potential issues in interpretation of the visualizations.

C6: Story Telling

The story I want to tell with my dashboard is that a comparison of patient demographics and condition prevalence between the WGU dataset and the CDC BRFSS dataset can reveal potential trends and disparities to inform targeted interventions that might improve patient care at WGU. The Patient Age Distribution bar chart displays the distribution of patients across different age categories for both the WGU dataset and the CDC BRFSS dataset. The x-axis represents age ranges (e.g., 18-24, 25-34, etc.) and the y-axis represents the number of patients in each category. By providing the demographic makeup of the two datasets, stakeholders may quickly assess whether WGU's patient demographics follow trends seen in the broader population or determine if there are notable deviations. For example, if the WGU dataset shows a significantly higher proportion of patients in the "80 or older" age category compared to the CDC dataset, WGU may benefit from tailoring its services and resources to support geriatric care.

The heat maps on the second tab show the prevalence of each health condition (asthma, diabetes, overweight, stroke) across different age categories for both the WGU and CDC datasets. The rows represent the different data sources for each condition, CDC and WGU, the columns represent the age categories, and the cells are colored according to the prevalence rate (with redder colors indicating higher prevalence). This visualization is important to reveal potential disparities in condition prevalence between WGU and the national average. For example, if the WGU heatmap shows a significantly higher prevalence of all conditions for both genders among patients aged 80 or older compared to the CDC. This reinforces the prior premise that the older demographic at WGU suffers from more health conditions than the CDC population. The visual nature of the heatmap allows for quick identification of areas where WGU's performance deviates from the national benchmark.

These two visualizations work together to provide a comprehensive view of the data. By comparing WGU's data to the CDC benchmark,

these visualizations empower stakeholders to identify problematic trends and areas where targeted interventions may be needed to improve patient care.

C7: Audience Analysis

My primary audience for this presentation is the hospital's executive leadership team, the stakeholders responsible for executive decision-making. Based on personal experience presenting data to executive teams, I aimed to create a simple but informative Tableau presentation that included all data of potential interest with intuitive visualization styles and controls. By including an introduction that explained the basis and purpose of the dashboards, as well as a brief overview of their general design, I attempted to address any initial questions without the need for intervention from the data analysis team. The summary on the final tab was also designed with this intention in mind, as I know that sometimes statistical visualizations can be intimidating or confusing for some audiences, so explaining the findings in layman's terms could help to facilitate digestion of the information.

I excluded most technical details from the Tableau presentation, as while these may be useful to those interested or directly involved in the data analysis process, they tend to be insignificant at the executive level, provided that the process succeeded in achieving the project's goal. The information is of course available upon request, just not included in the main presentation.

C8: Universal Access

To make my presentation as universally accessible as possible, I chose statistical visualizations that were relatively simple to interpret, such as bar charts and heatmaps, and avoided potentially more complex charts like scatter plots or box plots. I avoided combining multiple variables on the same axes, and instead gave users the option of incorporating additional layers to the analysis through built-in filters, e.g. gender.

Additionally, I made sure to include a textual summary on the final tab of the presentation to ensure correct interpretation of the information. In my experience, audience members can sometimes doubt their own interpretation of visualizations, so adding supportive text in this manner served to address this concern without requiring the data analysis team to step in and personally explain in person.

C9: Audience Engagement

Harvard Business School describes data storytelling as having three main components: data, narrative, and visualizations (Cote, 2021). I attempted to follow this advice by explaining my data sources and the purpose of the presentation in the introduction, essentially establishing the "plot" of the story to follow. The next two sections focused on interactive visualizations that engaged the audience by allowing them to selectively choose what details they wanted to focus on while also broadly covering all the main points of the presentation's purpose. The presentation ended with a narrative summary of the findings of the analysis as well as their implications for patient care and treatment at WGU.

D: Code References

Categorize age into another column age group. (2020, June 18). Stack Overflow. https://stackoverflow.com/questions/62446911/categorize-age-into-another-column-age-group

Pandas.DataFrame.rename — pandas 2.1.0 documentation. (n.d.). pandas - Python Data Analysis Library. https://pandas.pydata.org/docs/reference/api/pandas.DataFrame.rename.html

E: Source References

Cote, C. (2021, November 23). Data storytelling: How to tell a story with data. Business Insights Blog. https://online.hbs.edu/blog/post/data-storytelling

Kumar, K. (2025, May 4). BRFSS 2020 heart disease Dataset(Cleaned version). Zenodo. https://zenodo.org/records/15336526

Shaffer, J. (2016, April 20). 5 Tips on Designing Colorblind-Friendly Visualizations. Tableau. https://www.tableau.com/blog/examining-data-viz-rules-dont-use-red-green-together